

PARTNERSHIP PLAN

GAUL

GROUPE AÉROSPATIAL DE L'UNIVERSITÉ LAVAL

UNIVERSITÉ LAVAL

Conceptualized by Marguerite Ricard et Florence Boucher

Groupe aérospatial de l'université Laval



Our divisions

Aerostructure

- Design and manufacturing of the rocket's structural components, as well as the integration of the rocket's other systems.
- Materials used: composite materials such as fiberglass and carbon fiber.

Propulsion

- Development, design, and manufacturing of a liquid engine.
- Equipment and materials used: isopropanol, valves, nitrous oxide (N_2O), machining services, and instrumentation for measuring various parameters during testing

Payload

- Design and build of an onboard payload for the rocket. Development of a Geiger counter and integration of onboard cameras.
- Equipment used: 3D printer and 3D-printing plastics, electronic bays.

Avionics

- Design of electronic circuits for our rockets and for several of our ground-based electronic systems.
- Equipment used: PCBs, RF antennas, altimeters, onboard computers, GPS systems, telemetry, and arming systems.

Who we are

The Université Laval Aerospace Group (GAUL), founded in 1993, brings together around fifty students from the Faculty of Science and Engineering who work collaboratively on the design and construction of high-power rockets launched in both national and international competitions.

The many projects at GAUL provide our members with the opportunity to apply the knowledge gained throughout their academic journey while developing essential engineering and research skills such as teamwork, problem-solving, and project management.

Contact Information

gaul@fsg.ulaval.ca

https://gaulspace.web.app/home

https://www.instagram.com/gaul.ul/

https://www.facebook.com/groupeaerospatialul

http://www.youtube.com/@projetGAUL

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COMPETITIONS

Launch Canada

The first national rocket competition in Canada, designed to inspire and foster aerospace engineering talent. This event brings together university students and rocketry enthusiasts to drive innovation and advance the future of Canadian aerospace.



IREC



The International Rocket Engineering Competition (IREC) is an event where student rocketry teams from around the world compete by launching their rockets. The competition is open to student teams worldwide, including those from the United States.

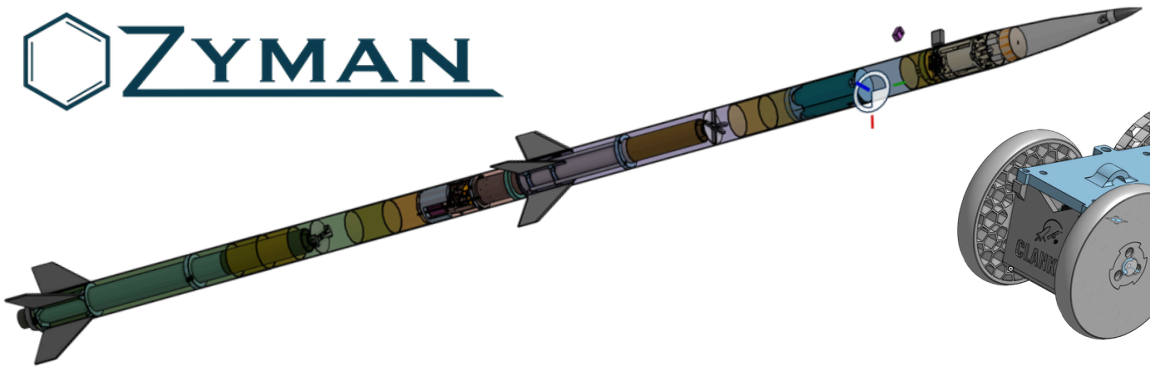
Far-Out

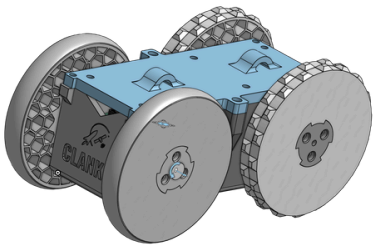
The FAR-OUT (Friends of Amateur Rocketry: Oxidizers Uninhibited Tournament) is a student rocketry competition focused on hybrid and liquid-propellant engines, giving university teams the opportunity to launch their rockets to target altitudes while carrying a payload.



OUR 2025-2026 ACHIEVEMENTS

ZYMAN



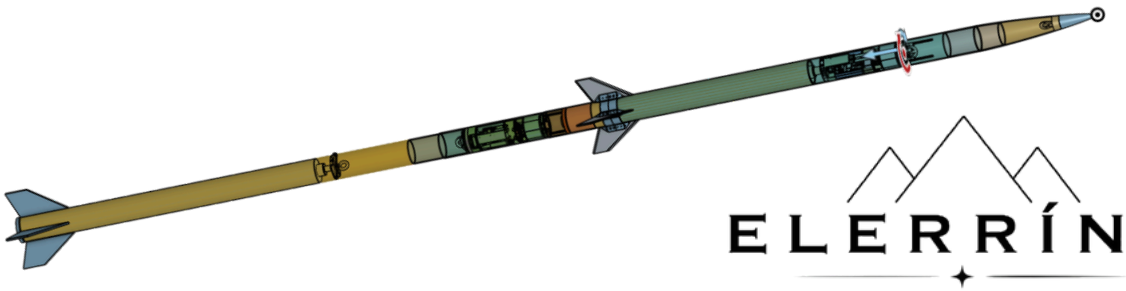




At IREC 2026, GAUL launched Ozyman, a two-stage rocket, in the 45,000-foot category. It was designed to reach Mach 1.6 and carry a 2 kg payload, including a rover.
The rover deployed successfully at the target altitude, and recovery was nominal. Unfortunately, the second stage, powered by our N1000, failed to ignite due to circumstances beyond our control. The rocket reached an apogee of 13,700 feet.

In 2026, GAUL took on a new challenge by designing and launching Ellerína, a two-stage rocket capable of reaching 97,000+ feet at Launch Canada. Building on the success of ALTERA, our first two-stage rocket, we set our sights on the Canadian altitude record for a solid-motor rocket.
Unfortunately, the second stage did not ignite: our safety systems prevented ignition beyond the 15° tilt limit. The good news is that our safety systems worked exactly as intended!
So while we didn't break the Canadian record, we still set a new GAUL altitude record of 33,800 feet by relaunching the second stage at the same competition.



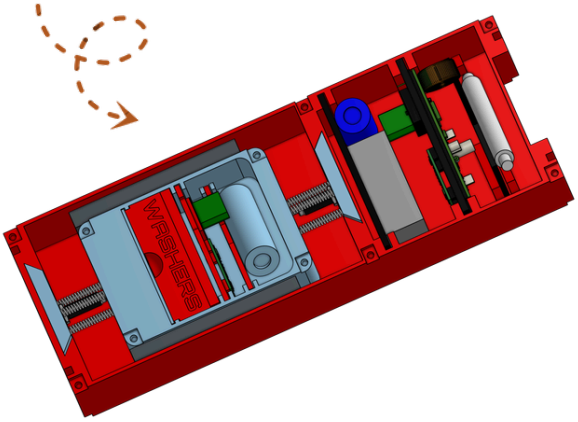


ELERRÍNA

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PROJECTS FOR 2026-2027

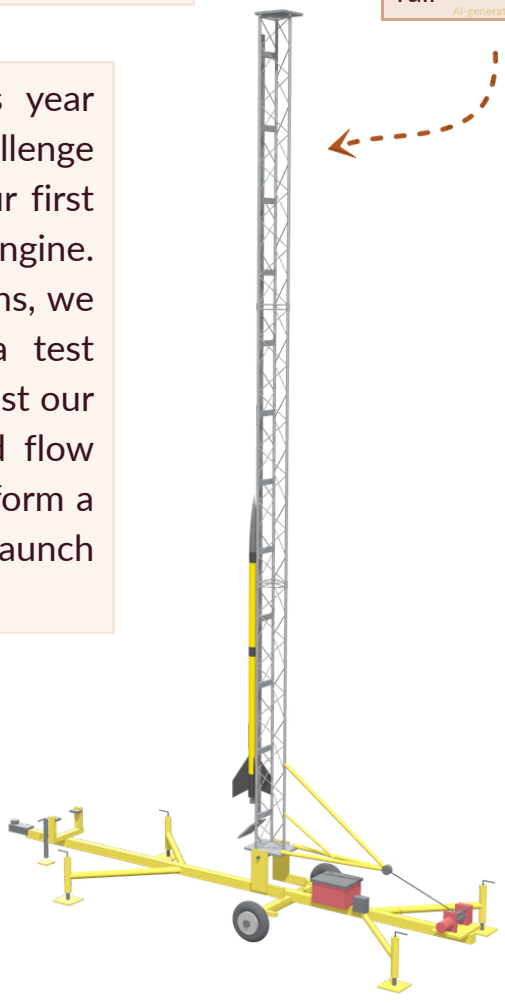
Modeling of the preliminary version of our liquid engine



Our payload division is currently working on equipping our rocket, to be launched at Launch Canada 2027, with a Geiger counter that will measure gamma rays at high altitude during flight. Vibration sensors will also track oscillations, all housed in a PLA or PETG enclosure.

One of our projects is to design a two-stage rocket capable of reaching 100,000 feet, which would break the Canadian amateur rocketry record of 67,000 feet. We will also build our own launch rail, roughly 30 feet tall, first to get a more stable rail than those provided at competitions, and second to ensure an off-rail velocity high enough for the rocket to be stable. It will be made of steel from DMX antenna tower sections fitted with a guide rail, and an electric winch will be used to raise it to vertical.

Here is a prototype of our launch rail



Our third project this year represents a new challenge for GAUL: it will be our first liquid-propellant engine. Over the coming months, we will be working on a test stand so that we can test our liquid engine in a cold flow test. The goal is to perform a hot fire test during Launch Canada 2027.



Modeling of the preliminary version of our liquid engine

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JOIN US !

The completion of all these projects requires significant resources. It would therefore be impossible for us to carry them out without the support of our valued partners. By joining our team through financial or material contributions, you will directly support the pursuit of our ambitions. In return, we commit to offering you privileged visibility during our competitions as well as at the numerous events we participate in each year.



PARTNERSHIP CATEGORIES

Benefits	Terrestrial Orbit 499 $ and less	Interplanetary 500 $ to 2499 $	Interstellar 2500 $ to 4999 $	Intergalactic 5000 $ and more
Logo on our website	X	X	X	X
Logo on competition shirts	X	X	X	X
Invitation to networking events		X	X	X
Logo on our rockets		X	X	X
Logo on promotional event material		X	X	X
Written description of your company at national events			X	X
Acknowledgment on our social media			X	X
Logo on posts during competitions				X
Mention during presentations				X
Written description of your company at international events				X

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